13/8/26

EXP NO:6

HAMMING CODE IMPLEMENTATION

Aim

write a code to implement error detection and correction using hamming code.

Procedure

- create sender program
- apply hamming code concept on binary data
- save output
- create receiver program
- it should read input from file
- apply hamming on binary data
- check for errors



Code

```
#include<stdio.h>
int encode(prtd){
    int d1 = (d>>0) &1;
    int d2 = (d >> 1)& 1;
    int d3 = (d>>2)& 1;
    int d4 = (d >>3) & 1;
    int p1 = d1^d2^d4;
    int p2 = d1 ^ d3 ^d4;
    int p4 = d2^ d3^ d4;
    return p1 |(p2<<1)|(d1 <<2)|(p4 <<3)|
        (d2<<4)|(d3<<5)|(d4 <<6)
}
int decode (int h, int * err){
    int b[8];
    for (int i=1; i<=7; i++){
        b[i] = (h >> (i-1))&1;
    }
    int c1 = b[1]^ b[3]^ b[5]^ b[7];
    int c2 = b[2] ^ b[3]^ b[6]^ b[7];
    int c3 = b[4] ^ b[5] ^ b[6]^ b[7];
    *err = (c3* 4)+(c2 * 2)+c1;
    return b[3]|(b[5]<<1)|(b[6]<<2)|
        (b[7]<<3)
```

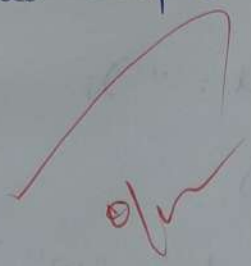
```
int main (){
    char str[50]; out[50];
    FILE * fp;
    printf("Enter text: ");
    scanf("%s", str);
    fp = fopen("channel", "w");
    for(int i=0; str[i]; i++){
        fprintf(fp, "%d %d", encode
                (str[i] & 0x0F)
    }
    fclose(fp);

fp = fopen("channel", "r");
int h1, h2, e1, e2, index = 0;
while(fscanf(fr, "%d %d", &h1, &h2)==2){
    int d1 = decode(h1, &e1);
    int d2 = decode(h2, &e2);
    if(e1||e2){
        printf("Error detected at bit position:
                %d %u", e1?e1 : e2);
    } else {
        out[index++] = d1 | (d2<<4);
    }
}
```

Output

enter text : HELLO

decoded output : HELLO



```
out [idx] = '\0';
fclose (fp);
printf ("decoded output text : %s \n", out);
return 0;
}
```

Result

Thus the hamming code is implemented successfully

13/8/22